CBCS SCHEME

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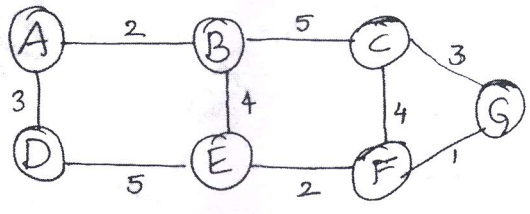
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BCS502

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2026
Computer Networks

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Data Communication. Identify five components of a data communications system. What are the three criteria's necessary for an effective and efficient network.	10	L2	CO1
	b.	Explain Fundamental characteristics of Data Communication, Categorize four basic topologies, cite and advantage of each type.	10	L2	CO1
		OR			
Q.2	a.	With a neat diagram, explain TCP/IP reference model.	10	L2	CO1
	b.	What are the three major classes of guided media? Enlist the advantages of optical fiber over twisted pair and co-axial cable.	10	L2	CO1
		Module – 2			
Q.3	a.	When a bit stream is transmitted using CRC coding with a generator polynomial $x^3 + x + 1$ for the data word $x^3 + 1$. Perform CRC encoding and decoding. Show the decoding case for corrupted and uncorrupted word.	10	L3	CO2
	b.	Explain the working of stop and wait protocol.	10	L2	CO2
		OR			
Q.4	a.	Explain with a neat diagram Transition phases of point-to point protocol.	10	L2	CO1
	b.	Illustrate Three Controlled Access Methods with neat representation. Design a scenario with 4-5 devices in a star/ring topology and demonstrate how token passing ensures orderly data transmission.	10	L3	CO2
		Module – 3			
Q.5	a.	Apply Dijkstra's algorithm to find the shortest path in a network given a topology as in the Fig Q5(a)  Fig Q5(a)	10	L3	CO2
	b.	Analyze the performance of a packet switched network during high traffic load. How does packswitching handle network congestion in comparison to circuit switching?	10	L4	CO3

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		OR			
Q.6	a.	Write a Pseudcode for Distance Vector Routing (DVR) Algorithm, which enables nodes in network to update their routing tables. Demonstrate how the each node determines the shortest path.	10	L3	CO2
	b.	Analyze the structure of an IPV4 data gram header, particularly focusing on fields like TTL-Time to Live and Checksum. Explain each field purpose.	10	L4	CO2
		Module – 4			
Q.7	a.	Explain the TCP segment structure and the significance of its key fields. Analyze how these fields are used to manage and track data segment during transmission.	10	L3	CO2
	b.	Compare and contrast UDP and TCP interms of Reliability and use cases.	10	L3	CO2
		OR			
Q.8	a.	Describe the process of TCP connection establishment and termination, including the three-way handshake. Analyze the situation if one of handshake packet is lost or delayed.	10	L4	CO2
	b.	Discuss the flow control and congestion control mechanism used in TCP. Illustrate how TCP adapts to prevent packet loss also handles the data errors.	10	L3	CO2
		Module – 5			
Q.9	a.	Discuss the importance of security in Application layer protocols. Explain common security measures implemented in protocol such as HTTPS, SMTPS with diagrams.	10	L2	CO3
	b.	What is Domain Name System (DNS)? Discus message structure and working mechanism with suitable diagram.	10	L2	CO2
		OR			
Q.10	a.	Explain the basic model of the File – Transfer Protocol (FTP). Describe its architecture and operations.	10	L2	CO1
	b.	Discuss the architecture of electronic mail systems. Explain the roles of SMTP, POP3, and IMAP protocols.	10	L2	CO2

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BCS502

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Computer Networks

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain fundamental characteristics and data representation in data communication.	6	L1	CO1
	b.	Discuss types of connection and the basic topologies in networks.	6	L1	CO1
	c.	Explain packet switching and circuit switching with neat diagrams.	8	L1	CO1
		OR			
Q.2	a.	Explain the layers of TCP/IP protocol suite.	8	L2	CO2
	b.	Explain types of packet switched networks and evaluate the total delay time of both.	12	L2	CO2
		Module – 2			
Q.3	a.	Explain types of errors and Hamming distance. Find the Hamming distance between the following : i) d(000, 011) ii) d(10101, 11110).	8	L3	CO2
	b.	Describe the working of CRC encoder and decoder. Perform division with respect to the following : Data word : 1001 Divisor : 1011.	12	L3	CO2
		OR			
Q.4	a.	Explain stop-and-wait protocol with FSM.	6	L2	CO2
	b.	Explain the three types of frames in HDLC.	8	L2	CO2
	c.	Discuss controlled-access protocol using reservation method.	6	L2	CO2
		Module – 3			
Q.5	a.	Explain the services offered by network layer.	6	L2	CO2
	b.	Define address space. Differentiate between classful addressing and classless addressing.	8	L2	CO2
	c.	Explain Network Address Resolution (NAT) with a neat diagram.	6	L2	CO3

		OR			
Q.6	**a.**	Explain IPv6 packet format in detail.	**6**	**L2**	**CO2**
	b.	Discuss D-V routing highlighting the importance of distance vector.	**7**	**L2**	**CO2**
	c.	Describe BGP protocol in detail.	**7**	**L2**	**CO4**
		Module – 4			
Q.7	**a.**	Explain the concept of port numbers mentioning ICANN ranges.	**5**	**L2**	**CO3**
	b.	Explain Go-Back-N protocol.	**9**	**L2**	**CO4**
	c.	Explain TCP segment format with a neat diagram.	**6**	**L3**	**CO3**
		OR			
Q.8	**a.**	Discuss the connection establishment in TCP.	**8**	**L2**	**CO3**
	b.	Explain error control in TCP using acknowledgements.	**4**	**L2**	**CO3**
	c.	Discuss three algorithms for handling congestion in TCP.	**8**	**L2**	**CO3**
		Module – 5			
Q.9	**a.**	Discuss application layer paradigms with neat diagram.	**5**	**L2**	**CO3**
	b.	Explain the use of sockets in process-to-process communication.	**7**	**L2**	**CO3**
	c.	Discuss the connection types in HTTP along with formats of messages.	**8**	**L2**	**CO3**
		OR			
Q.10	**a.**	Explain POP and IMAP protocols.	**8**	**L2**	**CO4**
	b.	Discuss the applications of SSH protocol.	**4**	**L2**	**CO4**
	c.	Explain resolution in DNS.	**8**	**L2**	**CO3**

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CBCS SCHEME - Make-Up Exam
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BCS502

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Computer Networks

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain different data flow (simplex, half-duplex, and full-duplex) modes with suitable examples.	10	L4	CO1
	b.	With a neat diagram explain functionalities of TCP/IP reference model architecture.	10	L2	CO1
		OR			
Q.2	a.	Briefly explain types of Guided Media (Twisted pair, Coaxial cable, and Fiber optics) with suitable applications.	10	L2	CO1
	b.	Explain datagram networks with suitable diagrams.	10	L2	CO1
		Module – 2			
Q.3	a.	What are the types of errors, explain Hamming distance and minimum hamming with suitable examples.	10	L2	CO2
	b.	Define linear block coding and CRC algorithm with suitable examples.	10	L2	CO2
		OR			
Q.4	a.	Example HDLC protocol with suitable diagrams.	10	L2	CO2
	b.	Explain flow diagram for the CSMA/CD protocol.	10	L2	CO2
		Module – 3			
Q.5	a.	Explain virtual-circuit packet switched network with suitable diagram.	10	L2	CO3
	b.	Explain DHCP message format and option format with suitable diagrams.	10	L2	CO3
		OR			
Q.6	a.	Describe open-loop congestion control (prevention) and closed-loop congestion control (removal) in detail.	10	L2	CO3
	b.	Explain message format of RIP and its performance.	10	L2	CO3
		Module – 4			
Q.7	a.	Discuss difference between Go-Back-N and selective repeat protocols.	10	L2	CO4
	b.	Explain TCP three - way Handshaking process with suitable diagram.	10	L2	CO4
		OR			
Q.8	a.	Explain TCP segment format with suitable diagrams.	10	L2	CO4
	b.	Explain congestion avoidance additive increase with suitable diagram.	10	L2	CO4
		Module – 5			
Q.9	a.	Explain types of network application architectures with a suitable diagrams.	10	L2	CO5
	b.	Explain HTTP protocol formats of the request and response messages.	10	L2	CO5
		OR			
Q.10	a.	Write short notes on cookies.	10	L2	CO5
	b.	Explain components of SSH and its packet format with suitable diagram.	10	L2	CO5

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BCS502

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025
Computer Networks

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Data Communications. Explain the characteristics and components of Data communication with neat diagram.	10	L2	CO1
	b.	With neat diagram explain the Layers in the TCP /IP protocol suite.	10	L2	CO1
		OR			
Q.2	a.	Explain in detail the guided and unguided Media transmission with suitable diagram.	12	L2	CO1
	b.	Describe the working of Datagram network with suitable sketches	08	L2	CO1
		Module – 2			
Q.3	a.	With a neat sketch describe the working of simple protocol of Data Link Layer. Develop a program to implement a sliding window protocol in the data link layer.	12	L2	CO2
	b.	Illustrate the stop and wait protocol of DLL with an example.	08	L2	CO2
		OR			
Q.4	a.	Solve : i) In parity check if the dataword is 1011. What is the code word? What happens at receiver, if the receive word is a) 10011 b) 10110 c)01011 ii) Generate CRC for the dataword $x^3 + 1$ and the generator $x^3 + x + 1$. What happens if the received word is 1000110. iii) Generate checksum of list of five 4-bit number (7,11,12,0,6) and verify the same at receiver.	12	L3	CO2
	b.	Illustrate the working of CSMA/CA with a flow diagram	08	L2	CO2
		Module – 3			
Q.5	a.	Summarize the packet format of IPV6 datagram with suitable diagram.	10	L2	CO2
	b.	Develop an algorithm for Distance Vector Routing and explain the same.	10	L2	CO4
		OR			
Q.6	a.	Explain MOSPF with an example and suitable diagram.	10	L3	CO4
	b.	Develop algorithm for Link state Routing and explain the same.	10	L2	CO4
		Module – 4			
Q.7	a.	Illustrate the working of Go-back-N protocol with an example	12	L2	CO4
	b.	Explain connectionless aand connection oriented services in Transport layer.	08	L2	CO2
		OR			
Q.8	a.	Illustrate the connection establishment and termination in TCP/IP with suitable sketches.	12	L2	CO3
	b.	With sketch of TCP segment format, describe its field.	08	L2	CO3
		Module – 5			
Q.9	a.	Explain FTP and its two connections.	10	L2	CO3
	b.	Explain SMTP with diagram and the mail transfer phases.	10	L2	CO3
		OR			
Q.10	a.	Explain MIME and its header.	10	L2	CO3
	b.	Explain SSH and its components with neat diagram.	10	L2	CO3

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Computer Networks

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is data communication? List and explain characteristics and components of communication model.	06	L1	CO1
	b.	Define switching. Explain Circuit Switched Network and Packet Switched Network.	06	L2	CO1
	c.	With neat sketch, explain different layers of TCP/IP protocol suite.	08	L2	CO1
		OR			
Q.2	a.	What are guided transmission media? Explain twisted pair cable in detail.	06	L1	CO1
	b.	What is Virtual Circuit Network (VCN)? With neat diagram, explain three phases involved in VCN.	08	L1	CO1
	c.	Write a note on Encapsulation and decapsulation at Source Host for TCP/IP protocol suite.	06	L2	CO1
		Module – 2			
Q.3	a.	Define Redundancy. Explain CRC encoder and CRC decoder operation with block diagram.	08	L2	CO2
	b.	Distinguish between Flow Control and Error Control. Explain Stop and Wait Protocol.	08	L2	CO2
	c.	List and explain Control Fields of I-frames, S-frames and U-frames.	04	L2	CO2
		OR			
Q.4	a.	What is Hamming distance? With example, explain Parity Check Code.	06	L1	CO2
	b.	Define Framing. Explain character oriented framing and bit-oriented framing.	06	L1	CO2
	c.	With flow diagram, explain CSMA/CA.	08	L2	CO2
		Module – 3			
Q.5	a.	Explain virtual-circuit approach to route the packets in packet-switched network.	10	L2	CO3
	b.	Illustrate the working of OSPF and BGP.	10	L3	CO3
		OR			
Q.6	a.	Explain IPv6 datagram format.	10	L2	CO3
	b.	Write an Dijikstra's algorithm to compute shortest path through graph.	06	L1	CO3
	c.	Write a note on Routing Information Protocol (RIP) algorithm.	04	L2	CO3
		Module – 4			
Q.7	a.	Explain Go-Back-N protocol working.	10	L2	CO4
	b.	With neat sketch, explain three-way handshaking of TCP connection establishment.	10	L2	CO4

		OR			
Q.8	**a.**	With an outline, explain selective repeat protocol.	**10**	**L2**	**CO4**
	b.	List and explain various services provided by User Datagram Protocol (UDP).	**10**	**L2**	**CO4**
		Module – 5			
Q.9	**a.**	Briefly explain Secure Shell (SSH).	**10**	**L2**	**CO4**
	b.	Write a note on Request message and response message formats of HTTP.	**10**	**L2**	**CO4**
		OR			
Q.10	**a.**	With neat diagram, explain the basic model of FTP.	**04**	**L2**	**CO4**
	b.	Describe the architecture of electronic mail (e-mail).	**06**	**L3**	**CO4**
	c.	Briefly explain Recursive Resolution and Iterative Resolution in DNS.	**10**	**L2**	**CO4**

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